HRO TYPE-C-31-M-12 | USB FIT AND DATUM REVIEW

Exact Bread local lands + primary HRO body drawing. NO verified STEP model. This is a dimensioned envelope, not physical qualification.

Panel X across / Y down / Z into case. Carrier F.Cu X4.1; board front Z2.

HRO shell: 8.94 wide x 7.35 deep x 3.26 high (drawing). Nominal shell axis X5.73.

Mating face Z0.8, Y33.5. Native footprint anchor (69.25,103.475), rotation 0 degrees.

Datum anchors: front shell tab 2.60 from face; rear 6.77 drawing / 6.78 local.

Ambiguous drawing vertical 0.50/1.64/4.18/5.79 chain is NOT used to invent a PCB-edge datum.

Local pad row: 7.695 from mouth; local land length 1.45, not claimed manufacturer-dimensioned.

Old GCT land row was 7.355 from mouth: HRO moves row 0.34 rearward; shell slot centres stay.

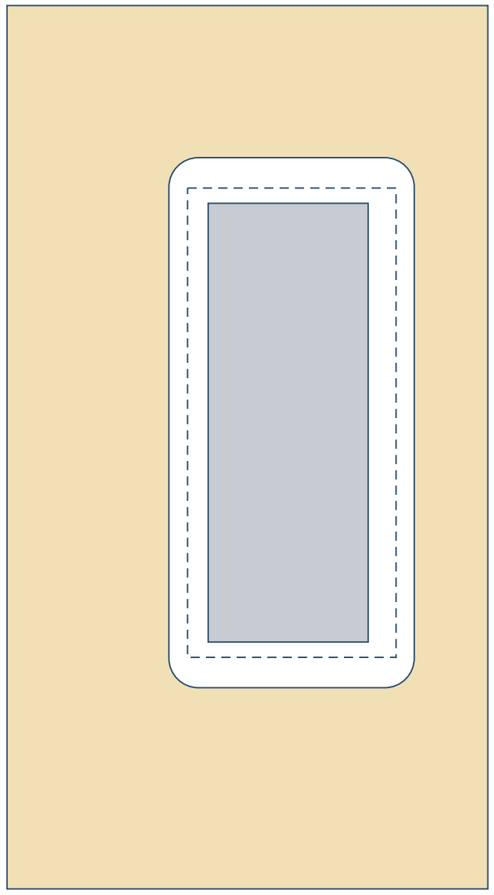
Local bosses: dia0.65 NPTH, nominal molded boss dia0.50; no peg shrinking.

Local shell drills: two 0.6 x1.7 rear and two 0.6 x1.2 front plated slots; no shell paste.

Front slot: nominal drill-to-board-edge 0.80; copper-to-board-edge 0.60.

A1/B12, A4/B9, A9/B4, A12/B1 share four wide PCB lands. S1 shell is GND.

A5=CC1, B5=CC2; all four data and two SBU contacts remain explicit NC.



FRONT VIEW: GRAY NOMINAL / DASHED WORST-CASE SHELL

Aperture: X3.30, Y28.10, W5.00, H10.80, R0.60 mm.

Delta from 024: X -0.93, Y -0.50, width +0.80, height +1.00.

Each machined wall +/-0.10; centre registration +/-0.10 X/Y.

Corner radius 0.50..0.70; profile acceptance THROUGH thickness.

BOTH finished boards 1.60 +/-0.16 mm, including coating.

Carrier back-face datum fixed: thickness error moves shell in X.

Assembly registration +/-0.10 X/Y; seating +0.05 X.

Shell width +0.15, height +0.10; all errors add worst-case.

Squareness <=0.5 deg; +/-0.02 tilt sweep through max1.76 panel.

Minimum modeled corner clearance: 0.233 mm.

In-plane angular registration <=0.5 deg; sweep +/-.040 X / .016 Y.

Assembly registration must be measured; M3 freeplay is not proof.

Panel9.8 x128.5 / carrier100.5 x30; D10/SW1/J7/brackets fixed.

Mouth recess 0.42..1.18; ahead of rear face 0.58..1.02 mm.

Cable insertion/overmould, ABS and actual assembled trial pending.

Bare-PCB aperture fit is NOT product/strength/bench qualification.

Fabrication process requirements: docs/fabrication.md.